

Supplementary Note: stability and support over the region-calling operating grid

S1 Motivation and what is being measured

The region-calling step has two nuisance parameters: the C-score threshold τ_C and the minimum region size $\ell_{\min}$. Reporting a single operating point invites the objection that the result was chosen; selecting the cell that maximises the number of significant regions would reintroduce, at the grid level, exactly the multiple-testing problem the region-level FDR controls within a cell. A natural response is to *integrate* over a grid $\mathcal{G}$ of operating points and report, for each locus, the fraction of cells in which it survives.

We evaluated this idea on the stickleback data and on the Nemo simulations. Four quantities are easily conflated, and we distinguish them throughout:

1. **Detection stability** — the fraction of grid cells in which a locus is still called as a region.
2. **Significance stability** — the fraction of cells in which the called region additionally passes the location-matched region-level test ($q_R < 0.05$ against the permutation null, Benjamini–Hochberg-adjusted within that cell).
3. **Genome-wide count FDR** — $\mathbb{E}[\#\text{null regions}]/\#\text{observed regions}$ at a cell, used to pick an operating point; not per-region.
4. **The location-matched p_R and q_R at one operating point** — the test that actually carries formal error control.

Quantities 1 and 2 are *robustness rankings*. Neither is a p -value, and neither confers false-discovery control across the grid; error control continues to come from quantity 4.

S2 The significance layer is redundant with detection

Across the full $25 \times 7 = 175$ -cell grid ($\tau_C = 0.02, 0.04, \dots, 0.50$; $\ell_{\min} \in \{1, 2, 3, 5, 10, 15, 20\}$), the 17 reported EMMAX regions, and four region-matching rules, the number of cells in which a region is *detected but not location-matched significant* is **zero**. Repeating the analysis on every null-restricted grid of Section S4 gives the same answer in all 19 grid $\times$ matching configurations tested: 0 discordant cells.

The mechanism is a resolution effect, not a biological one. With $B = 200$ permutations the empirical p -value floor is $1/(B + 1) = 0.00498$, and 27.8% of all region $\times$ cell combinations sit exactly at that floor because the among-population permutation seldom rebuilds an observed locus at its own position. Benjamini–Hochberg is a step-up procedure, so a set of k regions tied at the floor p_0 is rejected *en bloc* as soon as $p_0 \leq \alpha k/m$; ties rescue one another. Where $\ell_{\min}$ is large the cell tests

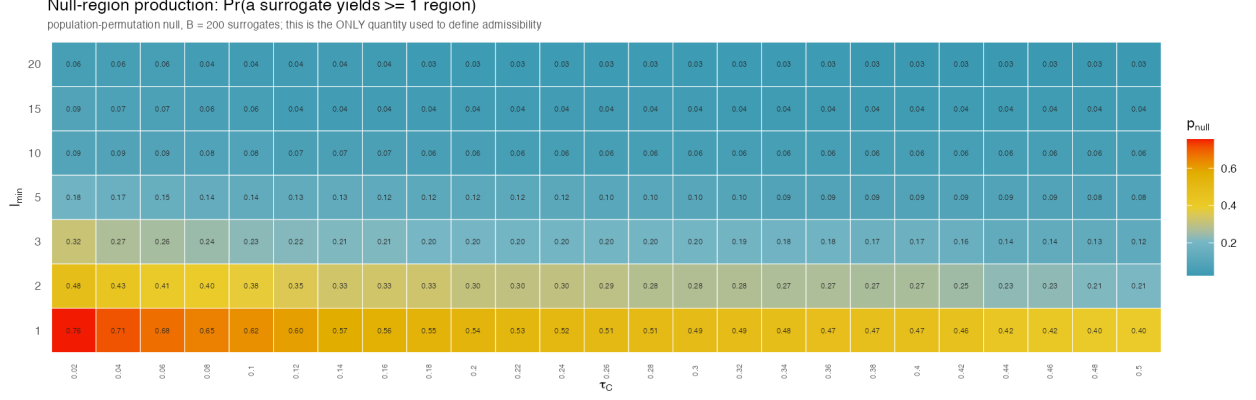


Figure S1: Probability that a permutation surrogate yields at least one region, over the $\tau_C \times \ell_{\min}$ grid ($B = 200$). The gradient runs almost entirely along $\ell_{\min}$; no cell reaches zero. Companion panels show the mean null region count (fig8), null marker coverage (fig9) and, on the identical grid, the observed region count (fig10).

only one to three regions and BH is trivially permissive. Either way, among detected cells the raw empirical p is already below 0.05 in 100% of cases and BH demotes none.

Consequence. On these data the “second-tier” score is numerically identical to detection stability. It should not be presented as a second inferential tier, nor described as integrating significance, and the colour scale of any figure built from it should be labelled as *support*, not significance.

S3 Null behaviour over the operating grid

We characterised the null directly: for every cell and every one of the $B = 200$ among-population permutation surrogates, regions were built with the same LD graph and the same region rules used for the observed data. Let $p_{\text{null}}(g) = \Pr_0(N_{\text{regions},g} > 0)$.

Three features matter (Fig. S1; Table S1).

There is no null-silent corner. p_{null} ranges from 0.025 to 0.755 and is **never zero** on any of the 175 cells. Some fraction of surrogates produces at least one LD-supported region everywhere in the parameter space.

The lenient axis is $\ell_{\min}$, not τ_C . Spearman correlation of p_{null} with $\ell_{\min}$ is -0.970 ; with τ_C it is only -0.216 . Across the entire τ_C range at $\ell_{\min} = 1$, p_{null} falls merely from 0.755 to 0.395 and never approaches any tolerance one might set. The “lenient corner” is in fact a horizontal band: any null-based restriction is close to a minimum-region-size requirement under another name.

There is no natural break. Sorted across cells, p_{null} has a median gap between adjacent values of 0.0000 and a largest gap of 0.045. It is a smooth continuum, so a cut point is *imposed* rather than discovered.

Table S1: Candidate null-only admissibility rules on the 175-cell grid. “CP” is the Clopper–Pearson upper confidence bound for p_{null} at $B = 200$. Every retained set is a single connected component and a contiguous high- τ_C suffix within each retained ℓ_{min} .

Rule	Cells retained	ℓ_{min} retained	Max p_{null} retained
$p_{\text{null}} \leq 0.20$	116	3, 5, 10, 15, 20	0.200
$p_{\text{null}} \leq 0.10$	86	5, 10, 15, 20	0.100
$p_{\text{null}} \leq 0.05$	42	15, 20	0.045
CP upper bound ≤ 0.10	61	10, 15, 20	0.055
CP upper bound ≤ 0.05	0	—	—
mean null coverage ≤ 0.001	107	all	0.470
mean null regions ≤ 0.5	81	5, 10, 15, 20	0.085

S4 Null-informed admissible grids

If no single operating point is to be chosen, one may instead let the null *exclude* cells that frequently manufacture spurious regions, and assess signals over what remains:

$$\mathcal{G}_{\text{adm}}(\epsilon) = \{g : p_{\text{null}}(g) \leq \epsilon\}.$$

Critically, this rule depends on null behaviour only; the empirical region burden was computed for diagnosis but never consulted in defining admissibility. (The two do covary strongly — Spearman 0.925 between p_{null} and the observed region count — which is precisely why the empirical side must be kept out of the rule.)

Finite B is the binding constraint. At $\epsilon = 0.05$ the point-estimate rule retains 42 cells but the confidence-bound rule retains **none**: not one cell on the grid can be *shown*, at $B = 200$, to have $p_{\text{null}} \leq 0.05$. At $\epsilon = 0.10$, 25 of the 86 retained cells (29%) are statistically indistinguishable from failing the rule. A cell observed at 20/200 has a Clopper–Pearson interval of $[0.062, 0.150]$, so $\epsilon = 0.05$ and $\epsilon = 0.10$ cannot be told apart with this null. Only $\epsilon = 0.20$ is reasonably resolved (13% ambiguous).

S5 Support of the reported regions across admissible grids

For a prespecified region r we report *detection support*

$$D_r = \frac{\#\{g \in \mathcal{G}_{\text{adm}} : r \text{ recovered in } g\}}{|\mathcal{G}_{\text{adm}}|},$$

with the denominator running over *all* admissible cells, including those in which nothing is detected. Regions are fixed *before* the grid search, as the region set at the reference point $(\tau_C, \ell_{\text{min}}) = (0.05, 3)$, and retained as member-marker vectors; a region is recovered in a cell when some called region returns at least half of its markers. Physical-span overlap is not used, because two physically overlapping components need not be the same LD-defined signal.

Table S2 shows that tightening the tolerance does not merely re-rank the regions. At $\epsilon = 0.05$ (and under the null-coverage rule) *all* 17 reported regions lose their support entirely. The reason is mechanical rather than evidential: nine of the seventeen regions contain four SNPs or fewer, so once admissibility requires $\ell_{\text{min}} \geq 10$ they can be recovered only by merging into a larger component. A filter that deletes the entire reported region set is not sharpening the inference.

Table S2: Detection support of the 17 reported regions under progressively stricter null tolerances. The reference point (0.05, 3) supplies the displayed boundaries only; it is not a grid cell and does not define admissibility.

Region	full grid (175 cells)	$\epsilon = 0.20$ (116)	$\epsilon = 0.10$ (86)	CP ≤ 0.10 (61)
Chr5:2.48–2.50 Mb	0.149	0.121	0.116	0.049
Chr1:21.50–21.93 Mb (inversion)	0.120	0.103	0.105	0.049
Chr20:0.18–0.26 Mb	0.120	0.103	0.105	0.049
Chr7:11.55–11.66 Mb	0.091	0.060	0.047	0.016
Chr4:12.81 Mb (<i>Eda</i>)	0.069	0.034	0.035	0.016
Chr12:16.52–16.55 Mb	0.074	0.034	0.012	0
Chr17:9.94 Mb	0.074	0.034	0.012	0
Chr4:27.55 Mb	0.171	0.026	0	0
Chr17:6.55–6.77 Mb	0.069	0.026	0	0
Chr20:8.89–8.90 Mb	0.051	0.026	0.023	0
Chr18:5.61 Mb	0.109	0.017	0	0
Chr4:22.51 Mb	0.143	0.009	0	0
Chr12:15.55 Mb	0.057	0.009	0	0
Chr1:27.28 Mb	0.063	0	0	0
Chr20:3.67 Mb	0.051	0	0	0
Chr3:13.03 Mb	0.034	0	0	0
Chr4:25.67–25.99 Mb	0.034	0	0	0
Regions with zero support	0/17	4/17	9/17	12/17
Spearman vs. full grid	1.000	0.654	0.404	0.477

Corroboration of the null rule. The regions that drop out are the ones the rule is meant to remove. Computed *after* the rule was fixed, and never used to construct it: the four regions losing all support at $\epsilon = 0.20$ are detected only in cells of mean $p_{\text{null}} = 0.488$, with 0% of their detecting cells admissible; the thirteen regions retaining support are detected in cells of mean $p_{\text{null}} = 0.361$, with 5–57% of their detecting cells admissible.

Tolerance-robust signals. Only three regions retain support under every tolerance examined: **Chr5:2.48–2.50 Mb**, the **Chr1 inversion**, and **Chr20:0.18–0.26 Mb**. *Eda* survives $\epsilon = 0.20$ but not the confidence-bound rule. All remaining regions are tolerance-dependent.

S6 Robustness of the summary itself

Matching rule. Requiring 25%, 50% or 75% of a region’s markers to be returned gives closely agreeing rankings (Spearman 0.776, 1.000, 0.972 against the 50% rule); a reciprocal variant that also caps how much larger the matched region may be gives 0.880. The retention threshold is not a sensitive choice. Permissive “any shared marker” or physical-overlap matching *is* different, because it lets a region be credited when it has been swallowed into a chromosome-scale component at low τ_C .

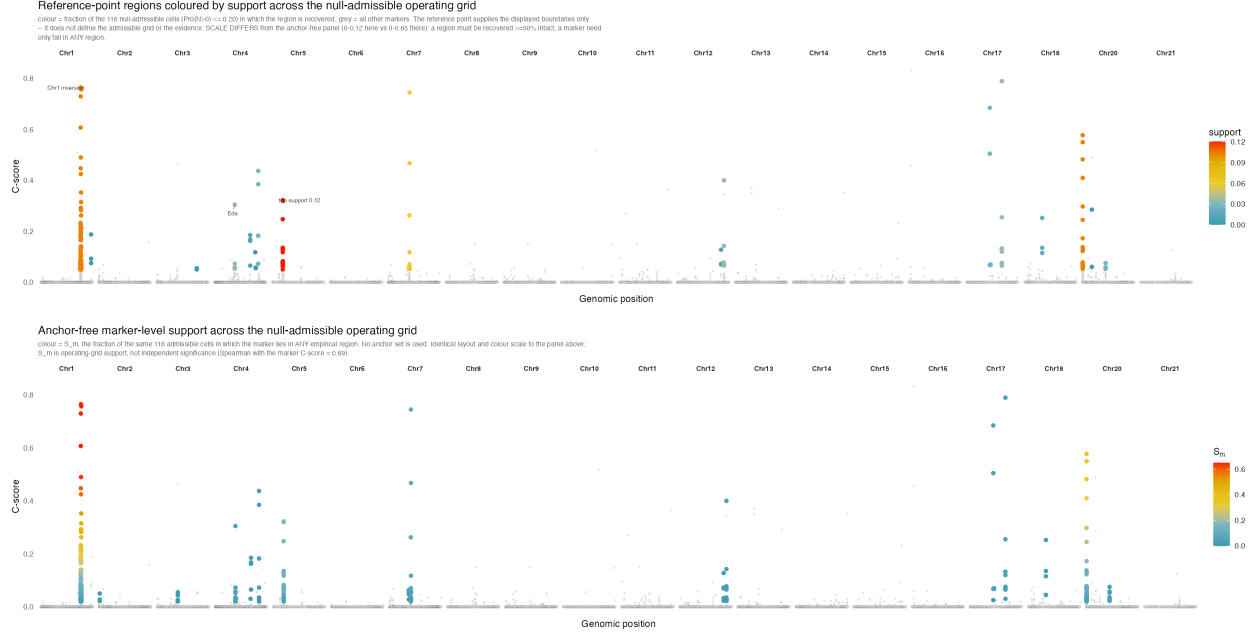


Figure S2: **Top:** the reported regions coloured by their support across the null-admissible grid ($\epsilon = 0.20$, 116 cells); grey, all other markers. **Bottom:** the anchor-free companion, every marker coloured by its own S_m on the same chromosome layout. The colour ranges differ (0–0.12 against 0–0.65) because a region must be returned at least half intact whereas a marker need only fall inside some region; a shared-scale version of the upper panel is provided for direct side-by-side reading. The colour is detection support, not significance (Section S2).

Reference point. Redefining the displayed regions at neighbouring points — (0.04, 3), (0.06, 3), (0.05, 2), (0.05, 5) — changes the inventory substantially (8 to 23 regions; 0 to 9 of them with zero support) but not the conclusion: Chr5:2.48–2.50 Mb is the best-supported region at four of the five reference points, with an identical maximum support of 0.121.

Grid range and density. Support is not invariant to grid design. Restricting to $\tau_C > 0.26$ sends every region to zero, whereas restricting to $\tau_C \leq 0.26$ reproduces the full-grid ranking exactly (Spearman 1.000): all of the information lives in the permissive half, and lengthening the τ_C range dilutes every score by inflating the denominator alone. A coarser but defensible grid (5×5) drives 13 of 17 regions to exactly zero.

Anchor-free check. Defining marker-level support S_m — the fraction of admissible cells in which marker m lies in any region — and grouping supported markers into families by the same r^2 linkage and 500 kb gap rule used for region calling, we find no signal that the reference point misses: on every admissible grid each family overlaps a reported region, region- and marker-level support agree at Spearman 0.992, and the largest S_m outside the reported set is 0.069. On the unrestricted grid 41 additional families appear, but with a median size of one marker. We stress that S_m correlates with the marker C-score at Spearman 0.69–0.85 and is therefore largely a re-expression of it; it is reported as support, never as independent evidence.

S7 Simulation check

On the Nemo simulations, where region truth is known, grid support does separate true from false regions: replicate-averaged over five environments, true regions score 0.437 against 0.094 for false ones (Wilcoxon $p \leq 7 \times 10^{-4}$ in every informative replicate). Ranking by support gives a precision-recall AUC of 0.901 ± 0.052 . However, ranking the same regions by a single number requiring no grid at all — the maximum marker C-score within the region — does better, 0.943 ± 0.025 . Detection and significance support are again identical to three decimal places. The simulations therefore validate the idea but do not motivate it over a far simpler statistic.

S8 What we claim, and what we do not

Safe. High support means a region is insensitive to the stated τ_C and $\ell_{\min}$ ranges. Low support means it lies near a calling boundary. The summary exposes parameter sensitivity and reduces reliance on any one cell. Reported support is meaningless without its grid, null basis, B , and matching rule, all of which we give.

Not claimed. That grid cells are independent, or that the fraction is a probability: they are strongly dependent, and $\tau_C \leq 0.26$ alone reproduces the full ranking. That integrating over the grid controls the false-discovery rate: it does not, and on these data it does not even add a second criterion beyond detection. That support is invariant to grid design, or that a null-derived admissibility boundary is a property of the data rather than of a chosen tolerance. That low support means a region is false.

Recommended reading of these results. Because p_{null} is a smooth continuum with no break, because the tolerance most would reach for ($\epsilon = 0.05$) is unresolvable at $B = 200$ and removes every reported region, and because the axis actually being filtered is $\ell_{\min}$ — already an explicit, reported parameter — we do not adopt a hard admissibility rule. We retain the single prespecified operating point with its location-matched q_R as the reported inference, and present the null landscape together with support at $\epsilon \in \{0.20, 0.10\}$ as a sensitivity analysis, noting that three regions (Chr5:2.48–2.50 Mb, the Chr1 inversion, Chr20:0.18–0.26 Mb) are robust across all tolerances examined.